COURSE DOCUMENTATION

Assignment No. 01
RPA Use Case: Employee Onboarding Analysis

Robotic Process Automation — Course Assignment

COURSE	ASSIGNMENT	DATE
Robotic Process Automation	No. 01	March 2026

DELIVERABLE
Single-page HTML Website + Documentation

1. Assignment Overview

This assignment required designing and documenting a comprehensive RPA use case analysis for a real-world HR business process. The deliverable is a single-page interactive website presenting the analysis across five structured sections: business process description, automation type evaluation, implementation planning, and projected benefits with data visualisations.

Deliverable Files

File	Type	Description
`rpa-onboarding.html`	HTML Website	Full single-page use case analysis with charts
`README.md`	Markdown	Plain-text documentation of assignment content
`README.pdf`	PDF	Formatted documentation (this file)

To view the website: open `rpa-onboarding.html` in any web browser (Chrome, Edge, Firefox, Safari).

2. Topic — Employee Onboarding Automation

HR onboarding was selected as the use case due to its high transaction volume, rule-based workflow structure, multi-system data entry requirements, and clear pain points — all characteristics that make it a strong candidate for RPA automation.

3. Website Structure & Content Summary

Section 1 — Introduction

Provides context on why HR onboarding is suitable for RPA. Covers the manual nature of existing processes, the cascade of activities triggered by each new hire, and the key characteristics — structured data, predictable rules, high volume — that make onboarding automation-ready.

Section 2 — Business Process Description

Documents the end-to-end onboarding workflow across seven steps:

1. **Job Offer & Acceptance** — HR sends offer letter via JD Edwards or manually; candidate confirms acceptance.
2. **Pre-Onboarding Setup** — Collect personal info, assign company email and credentials.
3. **Document Management** — Upload documents to OneDrive/SharePoint, verify completeness.
4. **Payroll & Benefits Setup** — Enter employee details into Paycon, enrol in standard benefits.
5. **IT & Equipment Provisioning** — Request laptop, phone, assign access to software systems.
6. **Orientation & Training Scheduling** — Schedule sessions, provide training links, assign LMS pathways.
7. **First-Day Follow-Up** — Confirm access to systems and document completion.

Systems Involved

JD Edwards — HRIS / ERP | Paycon — Payroll | Microsoft 365 — Email & IT | OneDrive / SharePoint — Document Storage

Key Pain Points

- ⚠ Manual re-entry of employee data across disconnected systems creates duplicate work and inconsistent records.
- ⚠ Time-consuming onboarding setup delays employee readiness and negatively impacts first-day experience.
- ⚠ Manual payroll entry introduces errors in compensation, tax configuration, and benefits enrolment (5–10% error rate).
- ⚠ Tracking document submissions and completeness via email is inefficient and prone to oversight.
- ⚠ JD Edwards, Paycon, and Microsoft 365 are not natively integrated, requiring manual data bridges.

3 (cont.) — Section 3: Use Case Evaluation

Recommended Automation Type: Hybrid RPA

A hybrid model is recommended — combining unattended automation for routine tasks with attended assistance for exception handling and human sign-off.

🤖 **Unattended Automation**

- Upload employee data to JD Edwards
- Enter payroll details into Paycon
- Create Microsoft 365 accounts
- Upload documents to SharePoint
- Generate standard payroll/benefits entries
- Trigger IT provisioning requests

👤 **Attended Automation**

- HR reviews flagged exceptions (visa workers)
- Approval of custom/non-standard benefits
- Manual adjustment of detected anomalies
- Edge-case document handling
- Compliance-sensitive sign-offs

✅ **Justification:** The majority of hires follow standardised, rule-based workflows — ideal for unattended automation at scale. Attended steps preserve human oversight for exceptions, balancing efficiency with compliance. Well-defined decision rules and structured data inputs reduce implementation risk.

4. Implementation Plan (High-Level)

Priority Automation Targets

1. **Employee data entry into JD Edwards** — highest volume, highest impact task.
2. **Microsoft 365 account creation** — bot provisions email, licence, and security groups by role template.
3. **Document uploads to SharePoint** — automate sorting, naming, uploading, and completeness validation.
4. **Payroll setup in Paycon** — replicate validated data from JD Edwards and configure benefits enrolment.

Tools & Platforms

RPA PLATFORMS

- UiPath
- Automation Anywhere
- Blue Prism
- Microsoft Power Automate

INTEGRATION METHODS

- JD Edwards REST API (preferred)
- Screen scraping (API fallback)
- Microsoft Graph API (M365)
- SharePoint REST API / Power Automate

Potential Challenges

- ⚠ Handling exception cases (visa holders, contract staff, custom benefits) without breaking the automated flow.
- ⚠ JD Edwards API limitations may require brittle screen-scraping, increasing maintenance burden.
- ⚠ Data security requirements for handling sensitive PII across automated workflows.
- ⚠ Inconsistent SharePoint/OneDrive permission structures may cause upload failures.
- ⚠ Change management — ensuring HR teams adopt the hybrid workflow rather than reverting to manual steps.

5. Projected Benefits (Quantified)

Metric	Before Automation	After Automation	Improvement
Time per hire (end-to-end)	**4–6 hours**	**1–2 hours**	~65–75% reduction
Payroll / HRIS error rate	**5–10%**	**<1%**	~90% reduction
Weekly HR hours (50 hires/week)	**250–300 hrs**	**50–75 hrs**	~75–80% savings
Document completeness rate	**~70–80%**	**>95%**	Significant compliance gain
Time-to-productivity (new hire)	**3–5 days**	**1–2 days**	Faster ramp-up
Cost / resource savings	**High manual labour cost**	**Reduced FTE overhead**	Significant cost avoidance

The website includes two interactive bar charts visualising these improvements:

- Chart 1 — Time per hire and weekly HR hours (before vs. after)
- Chart 2 — Payroll error rate and document completeness (before vs. after)

Summary: Hybrid RPA automation reduces manual HR onboarding effort by **70–80%**, brings error rates below **1%**, and accelerates new hire readiness from days to hours — freeing HR professionals from repetitive data entry to focus on strategic and relationship-driven work.

6. Technical Notes

- Built as a **single self-contained HTML file** — no build tools or server required.
- Charts rendered using **Chart.js** (loaded via CDN).
- Fully responsive layout using **CSS Grid and Flexbox**.
- No external frameworks — plain HTML, CSS, and vanilla JavaScript only.
- Compatible with all modern browsers: Chrome, Edge, Firefox, Safari.

How to Open

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Double-click   rpa-onboarding.html
— or —
Drag and drop into any web browser window
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